Term Work PCS 252

INSTRUCTION:

1. ***Use one side ruled sheets (Thick) for writing and DMP sheets for printouts.***
2. ***For question no. 1-13, every solution must contain the following: - Problem statement, Assumptions, Algorithms, Flowchart (back of ruled sheet), Printout of program and all possible outputs on separate sheets.***
3. ***For question no. 14-16, every solution must contain the following: - Problem statement, Assumptions, Algorithms, Printout of program and all possible outputs on separate sheets.***
4. ***Margin should be drawn on flowchart sheet 2cm from right and 1cm from rest of the three sides.***
5. ***Write your name, University roll no. and section with problem statement at the top of every code in the comment section.***

1. Write a C program to find roots of quadratic equation.
   Sample Input: Enter coefficients a, b and c: 4 1 0
   Sample Output: Roots are real numbers..... Roots are: 0.00 and -0.25

2. Write a C program to input a character and find whether it is number, alphabet or special symbol.

3. Write a C program to print the Fibonacci Series upto N Terms using recursion.

4. Write a C program to find the sum of the following series:

   a) $1^2 - 2^2 + 3^2 - 4^2$……upto n terms.
5. Write a C program to print the following pattern
   a) A
      B C
      D E F
      G H I J
      K L M N O
   b) 1
      0 1
      1 0 1
      0 1 0 1
      1 0 1 0 1

6. Write a C program to insert n elements in an array and replace each element of the array by its reverse.

7. Write a C program to check whether two strings are anagrams of each other. Both strings must contain exactly the same characters with the same frequency. Print “Valid Anagram” or “Not Anagram”.

8. Write a C program to input a square matrix and check whether it is a symmetric matrix or not. If symmetric, print “Matrix is Symmetric”, otherwise print “Matrix is Not Symmetric”.

9. Write a C program to input a matrix of order m x n and print its saddle point. If there is no saddle point then print a message no saddle point.(**Note:** A saddle point in a matrix is an element that is simultaneously the minimum in its row and the maximum in its column.**)**

10. Write a C program to store a predefined email ID using static initialization. Accept an email ID from the user and compare both using pointers. The comparison must be case insensitive. Print "Valid Email" or "Invalid Email".

11. Write a C program to input 'n' integers using dynamic memory allocation. Insert a new element at a position specified by the user and display the updated array.

12. Create a structure named "Book" having fields: Book_ID, Title, Author and Price. Accept details of 'n' books and sort the records in descending order of price and display the sorted list.

13. Create a structure named "Employee" to store employee details: Employee ID, Name, Department and Salary.
    (a) Write a function to display employees earning salary above a given amount.
    (b) Write a function to search employee details using Employee ID.

14. Write a C program to read 'n' integers from a file named "numbers.txt". Separate them into two different files prime.txt such that all prime numbers are copied in prime.txt and all non_prime numbers in non_prime.txt .

15. Write a C program to read a file named "Input.txt" and create another file "Output.txt" where all digits are removed and total number of vowels and consonants are counted and printed at the end of the file.

16. Write a C program to store details of 'n' products: Product name, Product ID, Quantity and Price. Store the data in a file "product.txt". Retrieve and display the records of products whose total cost is greater than 10,000 and store these records in another file "expensive.txt".